

Data experiment

Some fun: data experiment

There is no statistics without data.

- Hence, I ask you to provide some data as defined on the next slides.
- In order to set up an automatic process, it is very important that you follow the instructions carefully.
- Submission by **Tuesday, April 9, 6pm**

Later in the semester, I plan to provide some exercise in order to get bonus points for the exam.

I ask everyone to provide the following information:

1. A JSON file with some information about you.
2. A picture of yourself (or an avatar if you don't want to share your picture).
3. A picture of your favorite place.

I plan to assemble the pictures to a virtual group photo and use the provided information for the composition of possible groups for a (bonus) exercise.

Required data formats

- The file with the information about you should be in **JSON Format** (ISO/IEC 21778:2017), see <https://en.wikipedia.org/wiki/JSON>
- The file with your picture should be in **JPEG format** (ISO 10918-1) and have a size of 512×512 pixels, see <https://en.wikipedia.org/wiki/JPEG>
- The file with your favorite place should be in **JPEG format** (ISO 10918-1).

Needed data in JSON

The JSON file should contain the following fields:

Field	Description
Name	Your full name in your native language
Email	Your HTW email address
Country	Country of origin as an ISO 3166-1 Alpha-2 code
Languages	List of all languages you speak as ISO 639-1 codes. Use capital case for fluent, lower case for limited knowledge
Motto	Motto/aphorism characterizing you to write under your picture
Favorite place	Geographical coordinates of your favorite place worldwide specified in latitude & longitude following ISO 6709 code.
Source	State source of photo of your favorite place if it is not yours, otherwise add an empty string
Bonus exercise	State whether you will participate in the bonus exercise (yes / no)
Clearance	I herewith grant the lecturer the right to use and share the attached pictures for purposes related to this statistics course
Python skills	How would you rate your programming skills in Python (0-100)
Programming languages	In which languages do you have programming experiences? State as a list with elements from here: https://en.wikipedia.org/wiki/List_of_programming_languages
Operating system	State one computer operating system that you will use for statistics: Windows, Linux, Unix, Mac, Other, None

Example of a JSON file

```
{
  "Name": "Ralf Lenz",
  "Email": "ralf.lenz@htw-berlin.de",
  "Country": "DE",
  "Languages": ["DE", "ES", "EN", "fr", "pl"],
  "Motto": "Don't worry about your difficulties with math.
I can assure you that mine are even greater.",
  "Favorite place": [37.176601, 3.588],
  "Source": "",
  "Bonus exercise": "no",
  "Clearance": "I herewith grant the lecturer the right to use and share the attached
pictures for purposes related to this statistics course",
  "Python skills": 90,
  "Programming languages": ["Ada", "C", "C++", "Python", "PL/SQL", "GAMS", "MATLAB", "Maple"],
  "Operating system": "Windows"
}
```

Please submit the files as follows:

- **LastnameFirstname** should be the English transcription of your name
- The name of the JSON file should be **LastnameFirstname.json**
- The name of the file of your picture should be **LastnameFirstname.jpg**
- The name of the file of your favorite place should be **LastnameFirstname-place.jpg**
- All 3 files should be attached to the mail
- Send the email to ralf.lenz@htw-berlin.de
- The subject of the email should be:
Statistics: Data experiment for LastnameFirstname

Data submission is due to Wednesday, April 8, 6 pm

For the sake of simplicity, for the ones who already submitted data in the first round, let us only focus on the JSON files. For the others, still all three files are needed.

- There are a lot of errors of different types.
- To correct the errors, please read carefully the submission descriptions.
- Note that a **Source** of the photo is not given by a single word, but rather by the url.
- The subject of the email should be:

Statistics: Data experiment 2 for LastnameFirstname

Next submission is due to Friday, April 11, 6 pm

Data submission – Typical errors

1. Required email subject:

Statistics: Data experiment 2 for LastnameFirstname

Instead, many times the subject looks like this ...

- Fwd: JSON file XX
- Statistics: Data experiment for XX
- (no subject)
- Statistics: Data experiment 2 for LASTNAMEFirstname
- Data experiment 2 for XX
- Data experiment for XX
- Überarbeitete JSON Datei
- JSON Homework
- ...

2. Use your HTW mail

- If you have an HTW mail, but you sent the files from another mail, the submission will be ignored.
- Also use your HTW-mail as mail adress in the JSON file.

3. Send the required json file as a file of type .json

- Screenshots of json files will be ignored

Data submission – Typical errors

4. The json file should start and end with curly brackets
 - see my example on slide 6
5. Separate key-value pairs in the json files with commas
6. Check carefully the correctness of the keywords
 - Check for misspellings, e.g. *Favourite place* instead of *Favorite place*
 - Check lowercase and uppercase letters, e.g. *Python Skills* instead of *Python skills*
7. Did you write the name in your native language?
8. Did you use SO 3166-1 Alpha-2 code & ISO 639-1 codes for your country of origin and the languages
9. Important: the favorite place should contain a list of two numbers separated by comma
 - in particular, both numbers should not be stated as string
10. If you took the photo for your favorite place from the internet, state an entire url as source
11. Bonus exercise takes values: yes and no
 - values like 1, *true* should not be used
12. Did you state your python skills as a number between 0 and 100?
 - an empty string as value is not allowed

- Important: If the jpg file for your favorite place is $\geq 1000\text{KB}$, then compress it and resend it!

Next and last submission is due to Friday, April 18, 6 pm

The subject of the email should be:

Statistics: Data experiment 3 for LastnameFirstname